Overblik over Hesel ligningerne

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1 Standard form

HESEL ligningerne er givet på følgende form:

$$
\begin{aligned}
\mathrm{d}_t n + n\mathcal{K}(\phi) - \mathcal{K}(p_e) &= \Lambda_n \\
\mathrm{d}_t^0 \omega^* + \{\phi, p_i\} - \mathcal{K}(p_e + p_i) &= \Lambda_{\omega^*} \\
\frac{3}{2}\mathrm{d}_t p_e + \frac{5}{2} p_e \mathcal{K}(\phi) - \frac{5}{2}\mathcal{K}\left(\frac{p_e^2}{n}\right) &= \Lambda_{p_e} \\
\frac{3}{2}\mathrm{d}_t p_i + \frac{5}{2} p_i \mathcal{K}(\phi) + \frac{5}{2}\mathcal{K}\left(\frac{p_i^2}{n}\right) - p_i \mathcal{K}(p_e + p_i) &= \Lambda_{p_i}
\end{aligned}
$$

2 Lambda termer

Lambda termerne er givet ved:

$$
\begin{aligned}
\Lambda_n =& D_e \nabla_\perp^2 n - \frac{n}{\tau} - \frac{n - n_p}{\tau_p} - \alpha\left(\tilde{T}_e + \frac{\tilde{T}_e}{\bar{n}}\tilde{n} - \tilde{\phi}\right) \\
\Lambda_\omega^* =& D_i \nabla_\perp^2 \omega^* - \frac{\omega^*}{\tau} + \frac{\rho_s}{L_c}\left[1 - \exp\left(\phi_m - \frac{\phi_s}{T_{e,s}}\right)\right] \\
& - \alpha\left(\tilde{T}_e + \frac{\tilde{T}_e}{\bar{n}}\tilde{n} - \tilde{\phi}\right) \\
\Lambda_{p_e} =& \frac{5}{2} D_e \nabla_\perp^2 p_e + \left(\frac{16}{6} - \frac{5}{2}\right)\nabla \cdot (n \nabla_\perp T_e) - \frac{9 p_e}{2\tau} - \frac{T_e}{\tau^{SH}} - \Theta - \frac{p_e - p_{e,p}}{\tau_p} \\
& - \bar{T}_e \alpha\left(\tilde{T}_e + \frac{\tilde{T}_e}{\bar{n}}\tilde{n} - \tilde{\phi}\right) \\
\Lambda_{p_i} =& D_i \nabla_\perp^2 p_i - D_i T_i \nabla_\perp^2 n - \frac{9 p_i}{2\tau} + \Theta - \frac{p_i - p_{i,p}}{\tau_p} + p_i \Lambda_{\omega^*}
\end{aligned}
$$

3 Koeffecienter

Samling af koeffecienter:

$$T_{e,i} = p_{e,i}/n$$
$$D_i = \left(1 + \frac{R}{a} q^2\right) \frac{\rho_{i0}^2 v_{ii,0}}{\rho_{s0}^2 \omega_{i0}}$$
$$D_e = \left(1 + \frac{T_{i0}}{T_{e0}}\right) \left(1 + \frac{R}{a} q^2\right) \frac{\rho_{e0}^2 v_{ei,0}}{\rho_{s0}^2 \omega_{i0}}$$
$$\Theta = \frac{3m_e}{m_i} v_{ei} \left(p_e - p_i\right)$$
$$\tau = \frac{L_b}{2M_\| cs}$$
$$\tau^{SH} = \frac{L_b}{X_{\|,e}^{SH}}$$
$$\alpha = \frac{2T_{e0}}{v_{ei} m_e L_\|^2 \omega_{i0}}$$
$$c_s = \sqrt{(T_e + T_i)/m_i}$$
$$\phi_m = \ln(\sqrt{m_i/2\pi m_e})$$
$$\text{Ł}_\| = L_b = qR$$
$$\rho_s = \sqrt{\frac{T_{e0}}{m_i \Omega_{ei}^2}}$$

4 Forkotelser

De forkortede symboler er givet ved:

$$\omega^* = \nabla^2 \phi + \nabla^2 p_i$$

5 Operatorer

Herved er operatorerne defineret med:

$$\mathrm{d}_t^0 f = \partial_t f + \{\phi, f\}$$
$$\mathrm{d}_t f = \partial_t f + \frac{1}{B}\{\phi, f\}$$
$$\{f, g\} = \partial_x f \partial_y g - \partial_y f \partial_x g$$
$$\mathcal{K}(f) = -\frac{\rho_s}{R}\partial_y f$$
$$\bar{f} = \frac{1}{L_y}\int_0^{L_y} f \mathrm{d}y$$
$$\tilde{f} = f - \bar{f}$$
$$\nabla = (\partial_x, \partial_y, \partial_z)^T$$
$$\nabla_\perp = (\partial_x, \partial_y, 0)^T$$
$$\nabla^2 = \left(\partial_x^2, \partial_y^2, \partial_z^2\right)^T$$
$$\nabla_\perp^2 = \left(\partial_x^2, \partial_y^2, 0\right)^T$$

6 Funktioner

Funktionerne er gevet ved:

$$\begin{aligned} n &: n(x,y,t): \quad \text{"}Tæthed\text{"} \\ p_e &: p_e(x,y,t): \quad \text{"}Elektrontryk\text{"} \\ p_i &: p_i(x,y,t): \quad \text{"}Iontryk\text{"} \\ \phi &: \phi(x,y,t): \quad \text{"}Elektriskpotentiale\text{"} \end{aligned}$$

7 parametre

$$\begin{aligned} L_b = qR &\approx 8m \\ L_c &\approx 15m \\ T_{e0} &= 25eV \\ T_{i0} &= 25eV \\ M_\| &= 0.5 \\ 2M_\| c_s &= 50000ms^{-1} \\ c_s &= \sqrt{(T_e + T_i)/m_i} \end{aligned}$$

8 Variabler